

```
!pip install chromadb sentence-transformers
Requirement already satisfied: chromadb<0.4.15.2 in /usr/local/lib/python3.12/dist-packages (from chromadb) (0.4.15.2)
Collecting pypika>=0.48.9 (from chromadb)
  Downloading pypika-0.51.1-py2.py3-none-any.whl.metadata (51 kB)
    52.0/52.0 kB 1.2 MB/s eta 0:00:00
Requirement already satisfied: tqdm>=4.65.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (4.67.3)
Requirement already satisfied: overrides>=7.3.1 in /usr/local/lib/python3.12/dist-packages (from chromadb) (7.7.0)
Requirement already satisfied: importlib-resources in /usr/local/lib/python3.12/dist-packages (from chromadb) (6.5.2)
Requirement already satisfied: grpcio>=1.58.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (1.80.0)
Collecting bcrypt>=4.0.1 (from chromadb)
  Downloading bcrypt-5.0.0-cp39-abi3-manylinux_2_34_x86_64.whl.metadata (10 kB)
Requirement already satisfied: typer>=0.9.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (0.24.1)
Collecting kubernetes>=28.1.0 (from chromadb)
  Downloading kubernetes-35.0.0-py2.py3-none-any.whl.metadata (1.7 kB)
Requirement already satisfied: tenacity>=8.2.3 in /usr/local/lib/python3.12/dist-packages (from chromadb) (9.1.4)
Requirement already satisfied: pyyaml>=6.0.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (6.0.3)
Requirement already satisfied: mmh3>=4.0.1 in /usr/local/lib/python3.12/dist-packages (from chromadb) (5.2.1)
Requirement already satisfied: orjson>=3.9.12 in /usr/local/lib/python3.12/dist-packages (from chromadb) (3.11.8)
Requirement already satisfied: httpx>=0.27.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (0.28.1)
Requirement already satisfied: rich>=10.11.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (13.9.4)
Requirement already satisfied: jsonschema>=4.19.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (4.26.0)
Requirement already satisfied: transformers<6.0.0,>=4.41.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (5.0.0)
Requirement already satisfied: huggingface-hub>=0.20.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (1.8.0)
Requirement already satisfied: torch>=1.11.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (2.10.0+cpu)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (1.6.1)
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (1.16.3)
Requirement already satisfied: packaging>=24.0 in /usr/local/lib/python3.12/dist-packages (from build>=1.0.3->chromadb) (26.0)
Collecting pyproject_hooks (from build>=1.0.3->chromadb)
  Downloading pyproject_hooks-1.2.0-py3-none-any.whl.metadata (1.3 kB)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx>=0.27.0->chromadb) (4.13.0)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx>=0.27.0->chromadb) (2026.2.25)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx>=0.27.0->chromadb) (1.0.9)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx>=0.27.0->chromadb) (3.11)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx>=0.27.0->chromadb) (0.16.0)
Requirement already satisfied: filelock>=3.10.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->sentence-transformers) (3.25.2)
Requirement already satisfied: fspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->sentence-transformers) (2025.3.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->sentence-transformers) (1.4.2)
Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=4.19.0->chromadb) (26.1.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=4.19.0->chromadb) (2025.9.1)
Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=4.19.0->chromadb) (0.37.0)
Requirement already satisfied: rpds-py>=0.25.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=4.19.0->chromadb) (0.30.0)
Requirement already satisfied: six>=1.9.0 in /usr/local/lib/python3.12/dist-packages (from kubernetes>=28.1.0->chromadb) (1.17.0)
Requirement already satisfied: python-dateutil>=2.5.3 in /usr/local/lib/python3.12/dist-packages (from kubernetes>=28.1.0->chromadb) (2.9.0.post0)
Requirement already satisfied: websocket-client<0.40.0,!=0.41.*,!=0.42.*,>=0.32.0 in /usr/local/lib/python3.12/dist-packages (from kubernetes>=28.1.0->chromadb) (1.9.0)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from kubernetes>=28.1.0->chromadb) (2.32.4)
Requirement already satisfied: requests-oauthlib in /usr/local/lib/python3.12/dist-packages (from kubernetes>=28.1.0->chromadb) (2.0.0)
Requirement already satisfied: urllib3!=2.6.0,>=1.24.2 in /usr/local/lib/python3.12/dist-packages (from kubernetes>=28.1.0->chromadb) (2.5.0)
Collecting durationpy>=0.7 (from kubernetes>=28.1.0->chromadb)
  Downloading durationpy-0.10-py3-none-any.whl.metadata (340 bytes)
Requirement already satisfied: flatbuffers in /usr/local/lib/python3.12/dist-packages (from onnxruntime>=1.14.1->chromadb) (25.12.19)
Requirement already satisfied: protobuf in /usr/local/lib/python3.12/dist-packages (from onnxruntime>=1.14.1->chromadb) (5.29.6)
Requirement already satisfied: sympy in /usr/local/lib/python3.12/dist-packages (from onnxruntime>=1.14.1->chromadb) (1.14.0)
Requirement already satisfied: importlib-metadata<8.8.0,>=6.0 in /usr/local/lib/python3.12/dist-packages (from opentelemetry-api>=1.2.0->chromadb) (8.7.1)
Requirement already satisfied: googleapis-common-protos<=1.57 in /usr/local/lib/python3.12/dist-packages (from opentelemetry-exporter-otlp-proto-grpc>=1.2.0->chromadb) (1.73.1)
Collecting opentelemetry-exporter-otlp-proto-common==1.40.0 (from opentelemetry-exporter-otlp-proto-grpc>=1.2.0->chromadb)
  Downloading opentelemetry_exporter_otlp_proto_common-1.40.0-py3-none-any.whl.metadata (1.9 kB)
Collecting opentelemetry-proto==1.40.0 (from opentelemetry-exporter-otlp-proto-grpc>=1.2.0->chromadb)
  Downloading opentelemetry_proto-1.40.0-py3-none-any.whl.metadata (2.4 kB)
Collecting opentelemetry-sdk>=1.2.0 (from chromadb)
```

```
import chromadb
from sentence_transformers import SentenceTransformer
```

```
model = SentenceTransformer('all-MiniLM-L6-v2')
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret 'HF_TOKEN' does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
```

```
warnings.warn(
modules.json: 100% 349/349 [00:00<00:00, 32.9kB/s]
config_sentence_transformers.json: 100% 116/116 [00:00<00:00, 11.4kB/s]
README.md: 10,5k/? [00:00<00:00, 883kB/s]
sentence_bert_config.json: 100% 53,0/53,0 [00:00<00:00, 5,12kB/s]
config.json: 100% 612/612 [00:00<00:00, 55,5kB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
models.safetensors: 100% 90,9M/90,9M [00:01<00:00, 454MB/s]
Loading weights: 100% 103/103 [00:00<00:00, 441.77B/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
```

```
Key | Status |
-----+-----+
embeddings.position_ids | UNEXPECTED | |
```

```
Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
```

```
tokenizer_config.json: 100% 350/350 [00:00<00:00, 8,10kB/s]
vocab.txt: 232k/? [00:00<00:00, 8,65MB/s]
tokenizer.json: 466k/? [00:00<00:00, 20,2MB/s]
special_tokens_map.json: 100% 112/112 [00:00<00:00, 11,7kB/s]
config.json: 100% 190/190 [00:00<00:00, 18,8kB/s]
```

```
documents = [
    "Artificial Intelligence is the simulation of human intelligence.",
    "Machine Learning is a subset of AI that learns from data.",
    "Deep Learning uses neural networks.",
    "Natural Language Processing works with text data.",
    "RAG combines retrieval and generation.",
    "AI is used in healthcare and finance.",
    "Supervised learning uses labeled data.",
    "Unsupervised learning finds patterns in data.",
    "Reinforcement learning learns through rewards.",
    "Neural networks are inspired by the human brain."
]
```

```
embeddings = model.encode(documents).tolist()
```

```
client = chromadb.Client()
collection = client.create_collection(name="rag_lab")

for i, doc in enumerate(documents):
    collection.add(
        documents=[doc],
        embeddings=[embeddings[i]],
        ids=[str(i)]
    )
```

```
def retrieve(query, k=2):
    query_embedding = model.encode([query]).tolist()

    results = collection.query(
        query_embeddings=query_embedding,
        n_results=k
    )

    return results['documents'][0]
```

```
def rag_answer(query):
    retrieved_docs = retrieve(query)

    answer = " ".join(retrieved_docs)

    return answer, retrieved_docs
```

```
query = "What is Artificial Intelligence?"

answer, docs = rag_answer(query)

print("Question:", query)
print("\nRetrieved Docs:")
for d in docs:
    print("-", d)

print("\nFinal Answer:")
print(answer)
```

```
Question: What is Artificial Intelligence?

Retrieved Docs:
- Artificial Intelligence is the simulation of human intelligence.
- Machine Learning is a subset of AI that learns from data.

Final Answer:
Artificial Intelligence is the simulation of human intelligence. Machine Learning is a subset of AI that learns from data.
```

```
queries = [
    "Explain machine learning",
    "What is deep learning?",
    "Explain NLP"
]

for q in queries:
    answer, docs = rag_answer(q)

    print("\n=====")
    print("Question:", q)
    print("Answer:", answer)
```

```
=====
Question: Explain machine learning
Answer: Machine Learning is a subset of AI that learns from data. Supervised learning uses labeled data.

=====
Question: What is deep learning?
Answer: Deep Learning uses neural networks. Machine Learning is a subset of AI that learns from data.

=====
Question: Explain NLP
Answer: Natural Language Processing works with text data. Supervised learning uses labeled data.
```